

nRF24L01+ Quick Reference (for STM32 projects)

Pinout (Typical 8-pin Module)

Pin	Name	Function
1	GND	Ground
2	VCC	3.3 V supply (not 5V tolerant)
3	CE	Chip Enable, controls TX/RX mode
4	CSN	Chip Select (SPI active low)
5	SCK	SPI Clock
6	MOSI	SPI Master Out Slave In
7	MISO	SPI Master In Slave Out
8	IRQ	Interrupt (optional, active low)

SPI Interface

- Mode: SPI Mode 0 (CPOL=0, CPHA=0)
- Max clock: 10 MHz
- Active low CSN required for all SPI transactions

Power Requirements

- Supply: 1.9 – 3.6 V (typically 3.3 V)
- Current draw: ~11 mA RX, 9–12 mA TX (0 dBm)
- PA+LNA modules: up to 115 mA peaks
- Add decoupling capacitor (10–47 μ F) near VCC

Key Configuration Registers

Register	Description
CONFIG (0x00)	Enable CRC, power up, RX/TX mode
EN_AA (0x01)	Enable auto-acknowledge per data pipe
EN_RXADDR (0x02)	Enable RX addresses (pipes)
SETUP_AW (0x03)	Address width (3–5 bytes)
SETUP_RETR (0x04)	Auto retransmit delay and count
RF_CH (0x05)	RF channel (2400–2525 MHz)
RF_SETUP (0x06)	Data rate and TX power
STATUS (0x07)	IRQ flags and data ready indicators

RX_ADDR_Px	RX pipe addresses
TX_ADDR	TX address
FIFO_STATUS	TX/RX FIFO state

Air Data Rates

- 250 kbps: longest range, highest sensitivity
- 1 Mbps: balance of speed and range
- 2 Mbps: fastest, shorter range

STM32 Wiring Example

STM32 SPI1 to nRF24L01+:

- PA5 → SCK
- PA6 → MISO
- PA7 → MOSI
- PA4 → CSN
- PB0 → CE
- PB1 → IRQ (optional)